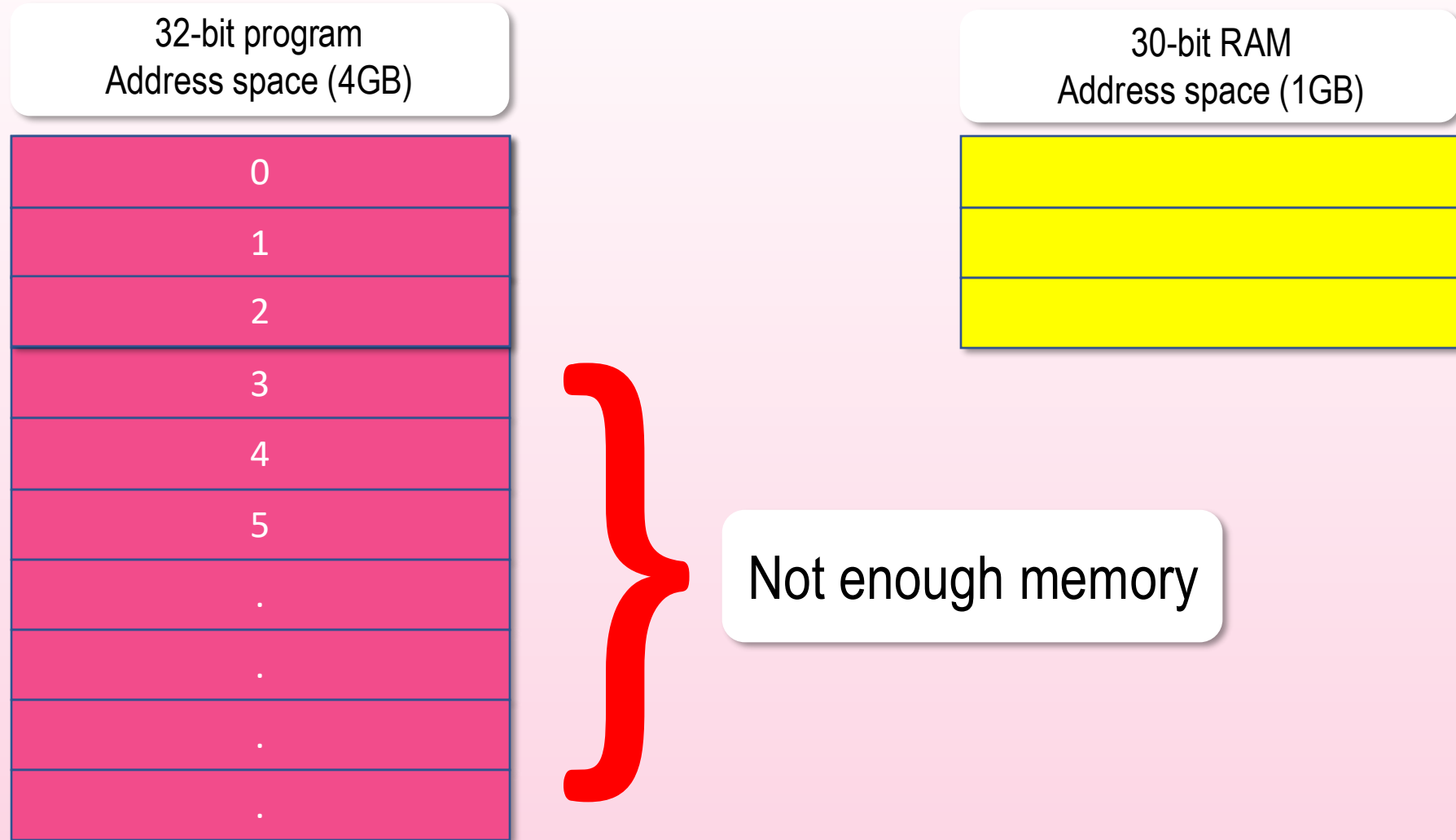


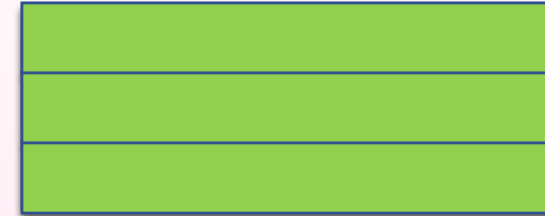
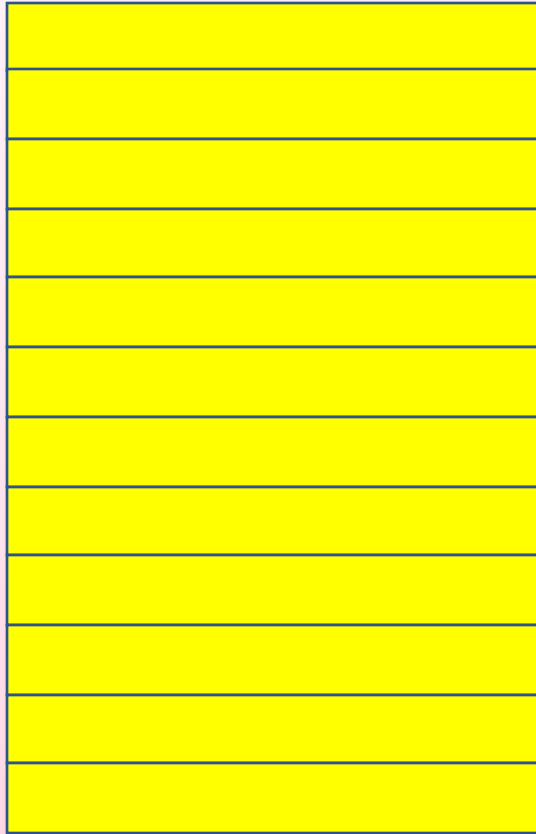
Virtual Memory And Disk

Problem : Not enough RAM

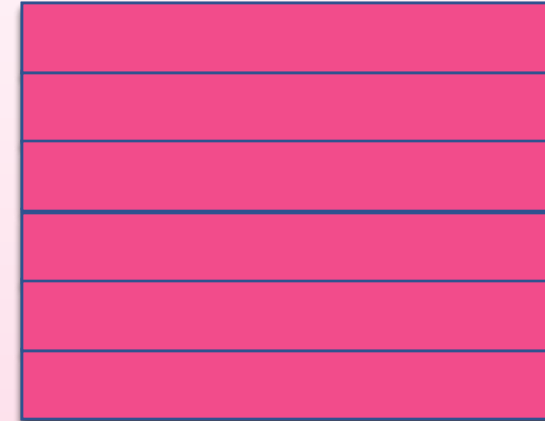


Problem : Contiguous memory allocation

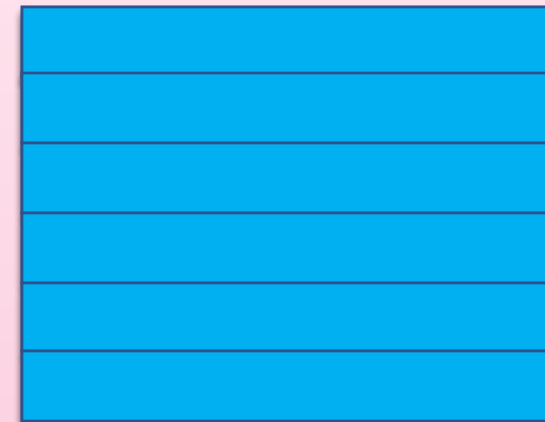
32-bit RAM
Address space (4GB)



Program 1
1GB



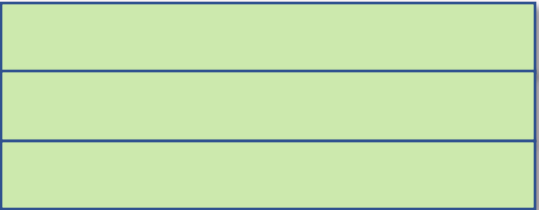
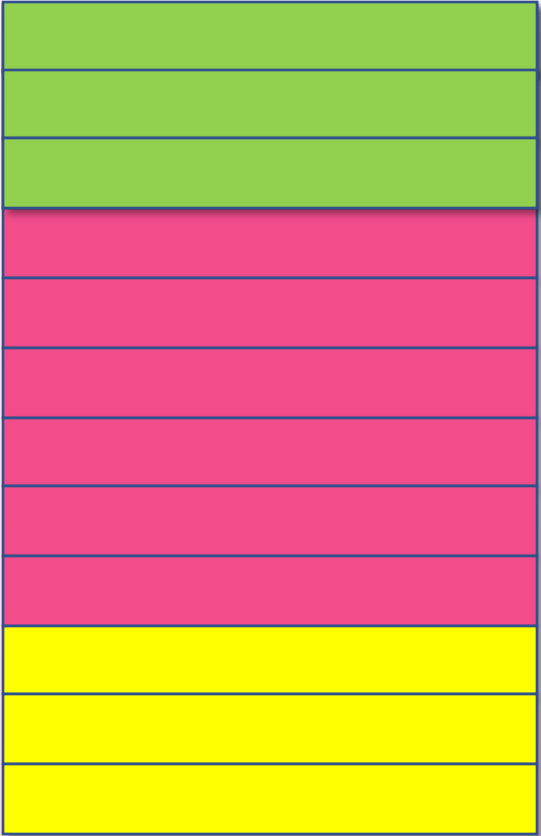
Program 2
2GB



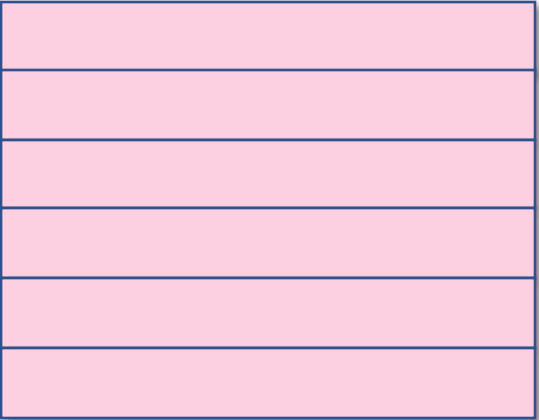
Program 3
2GB

Problem : Contiguous memory allocation

32-bit RAM
Address space (4GB)



Program 1
1GB

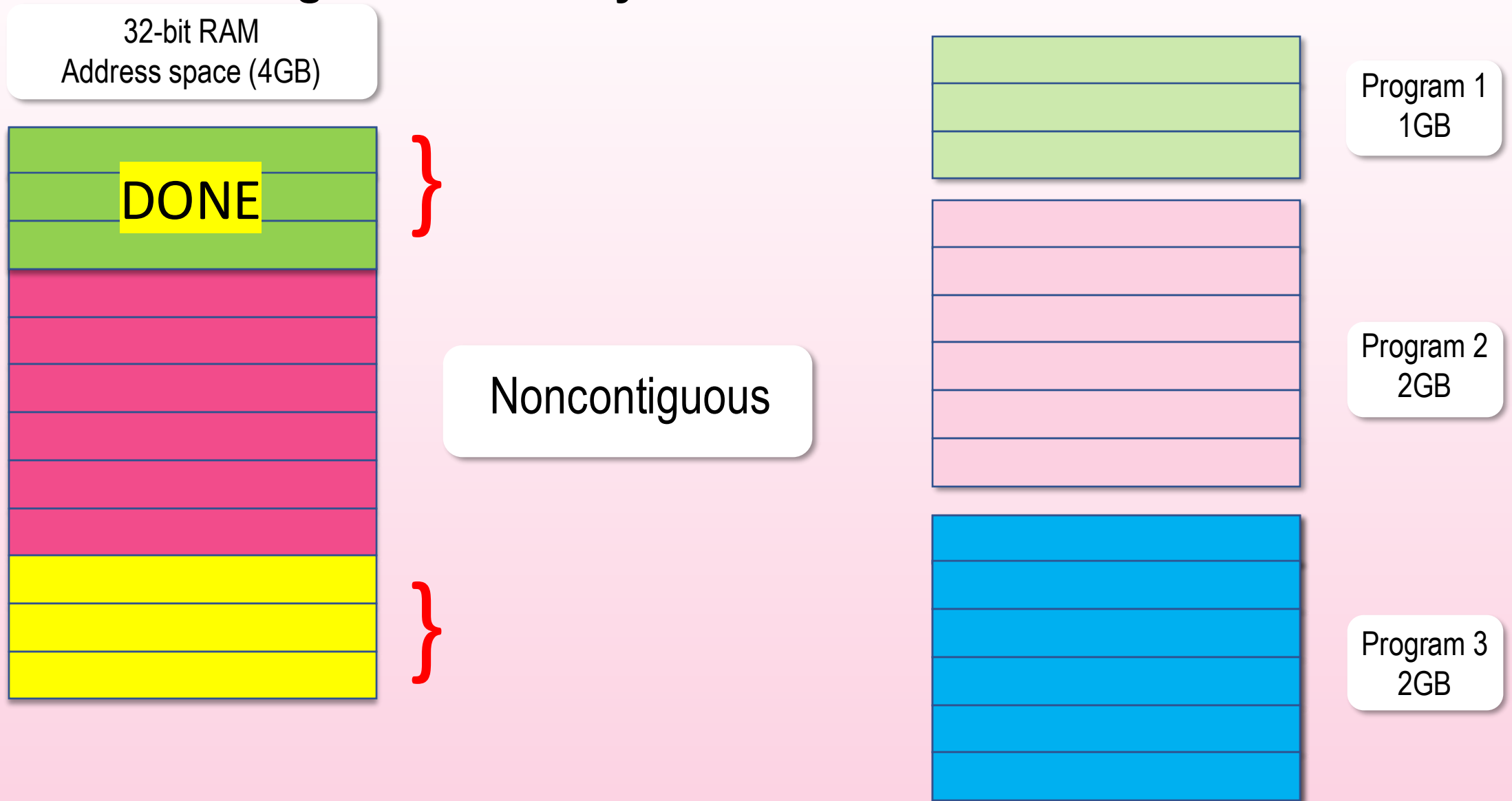


Program 2
2GB

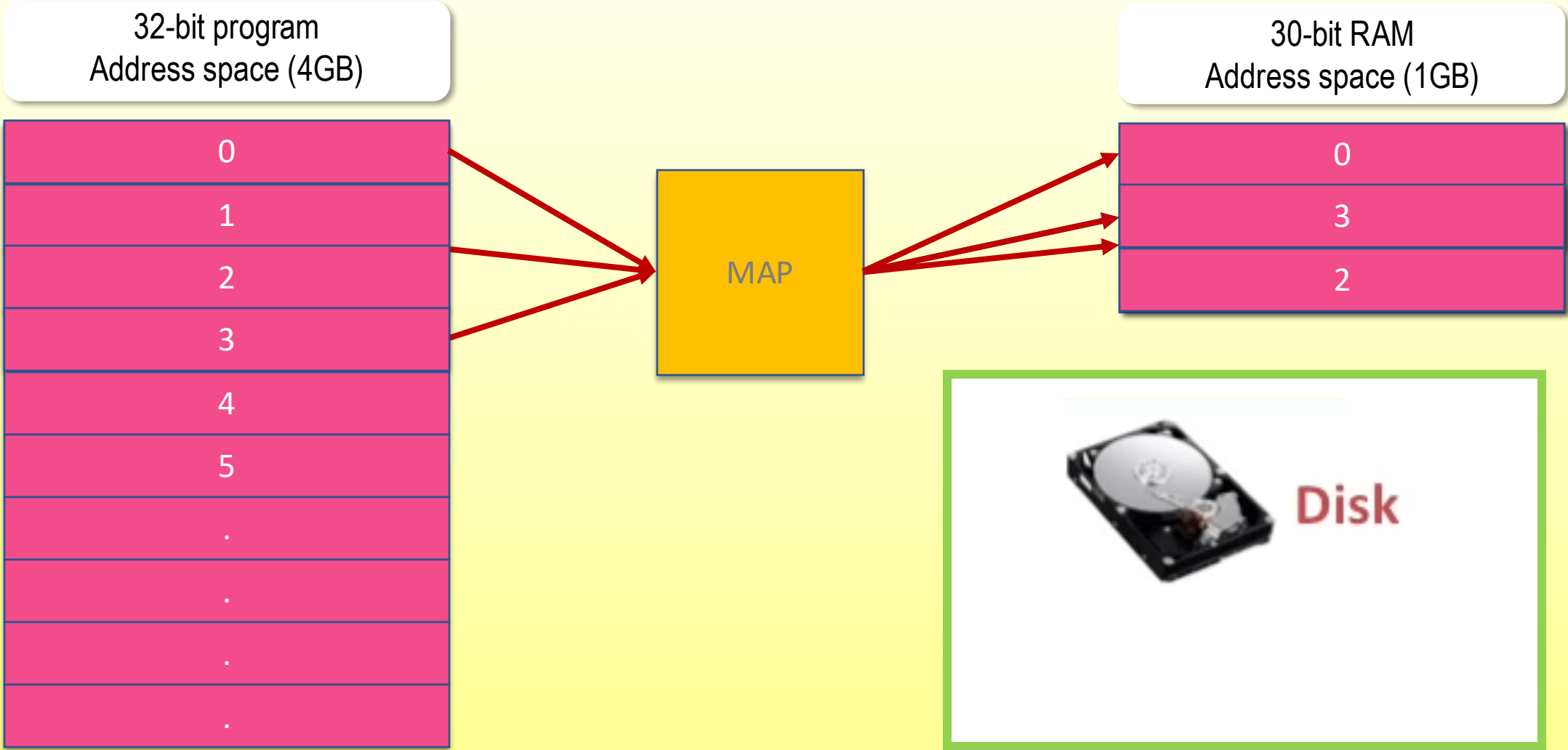


Program 3
2GB

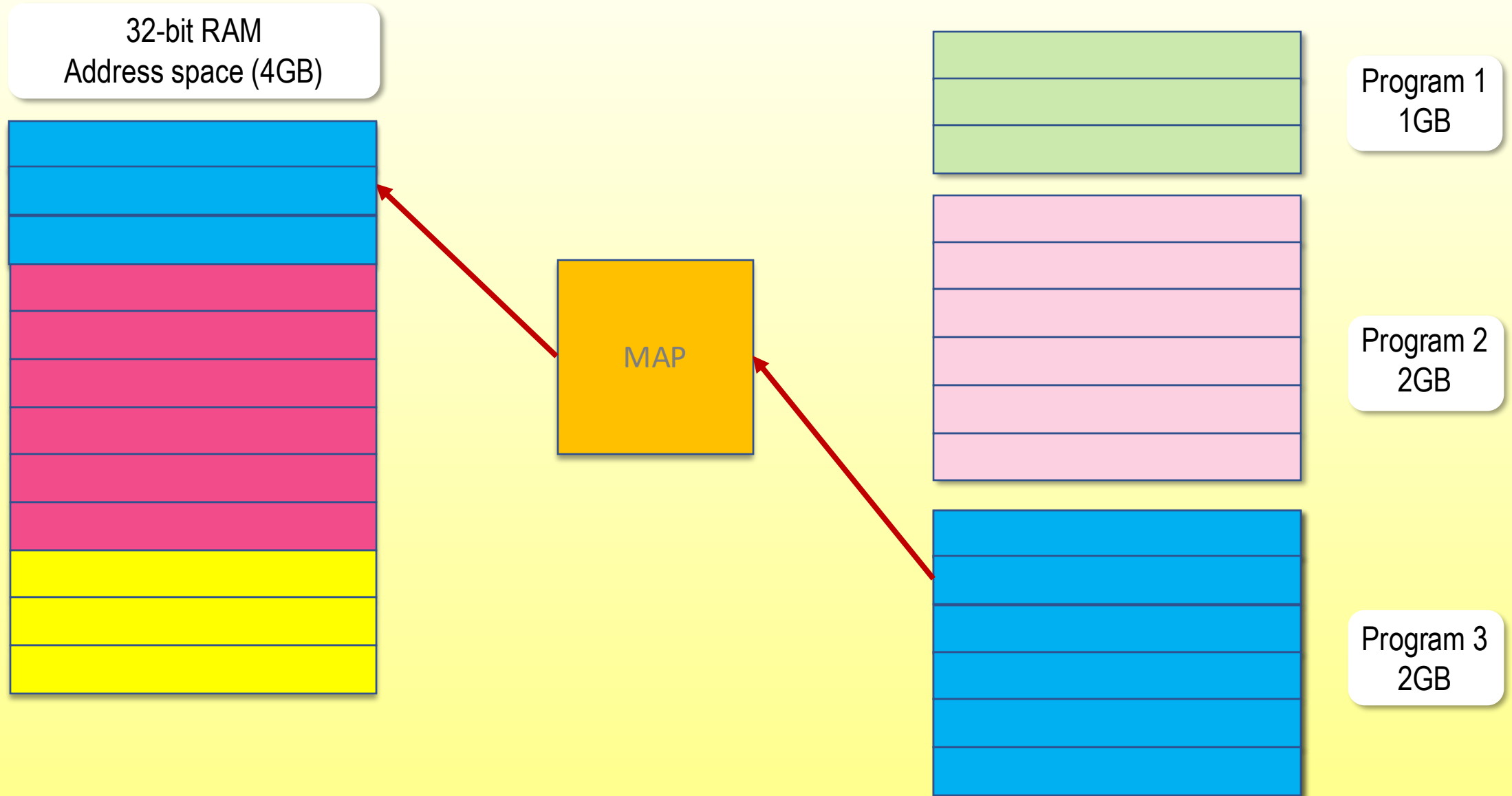
Problem : Contiguous memory allocation



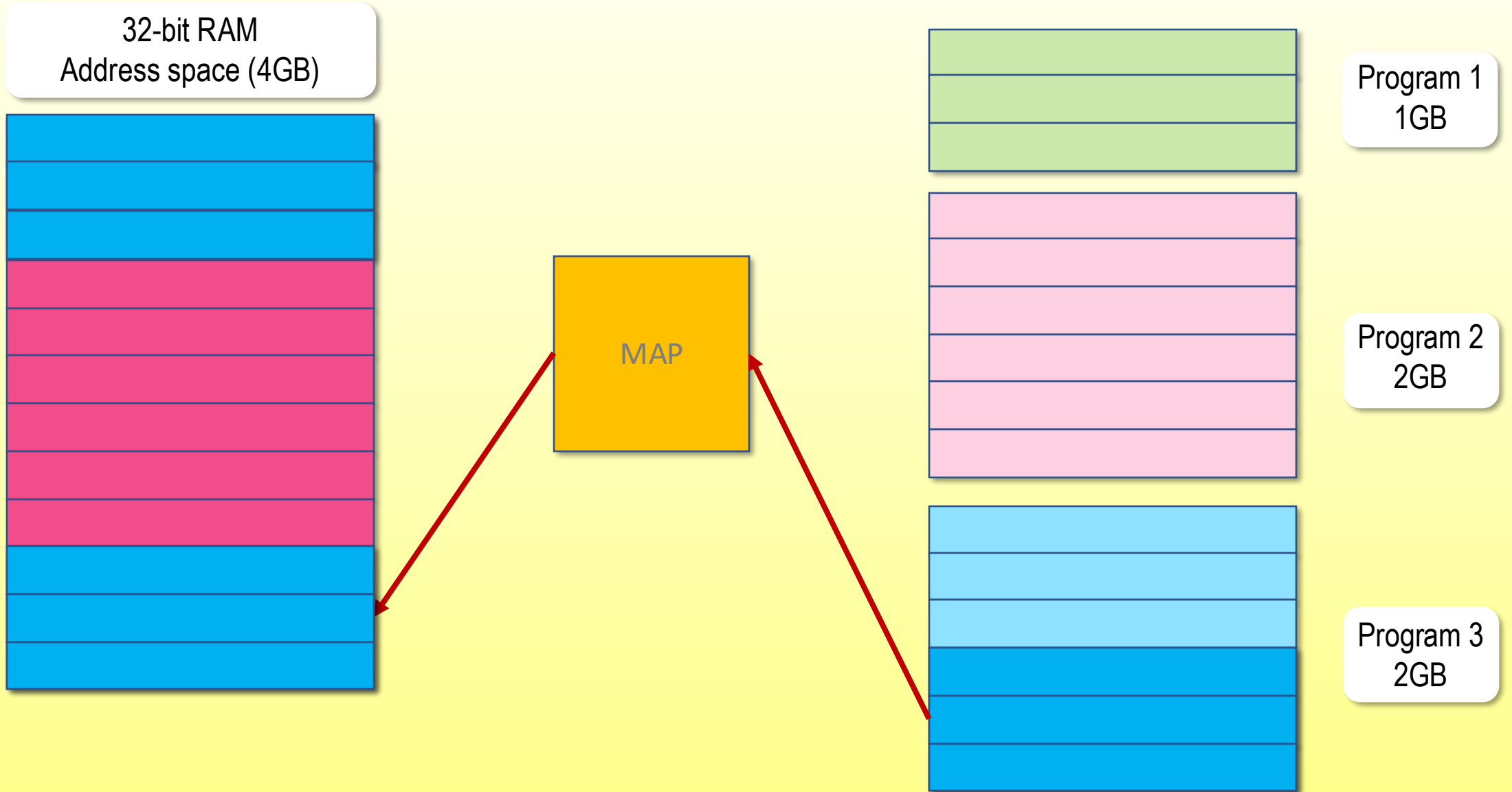
Solution : Not enough RAM



Solution : Contiguous memory allocation



Solution : Contiguous memory allocation



Address Mapping

Logical Address Space

Processor
ld R3, \$1024
ld R2, \$00FF
add R1, R2, R3
ld R4, \$0100
add R1, R1, R4

Address Mapping

MMU	
LA	PA
\$0100	disk
\$00FF	5
\$1024	3

Physical Address Space

RAM
0
1
2
3
4
5
6



Note: ပုံစံကုန်သော RAM ကိုလည်းကောင်း :: သို့ Virtual memory ကိုလည်းကောင်း disk မှာလည်းကောင်း

Page Mapping

Page Table
Translates VA to PA

MAP	
LA	PA
0-4095	4096-8191

4 row
(index # page
in vir)

2104 Page

ဥပမာ. 0-4095
was Virtual

map to

ဥပမာ. 4096-8191
was Physical

2104 Address

Address 0
was Virtual

map to

Address 4096
was Physical

Virtual Address
Space

16383

Page 3
4kB

12288

12287

Page 2
4kB

8192

8191

Page 1
4kB

4096

4095

Page 0
4kB

0

14-bit
2-bit : say page no 4
12-bit : say offset

Physical Address
Space

12287

4kB

8192

8191

4kB

4096

4095

4kB

0

4 page

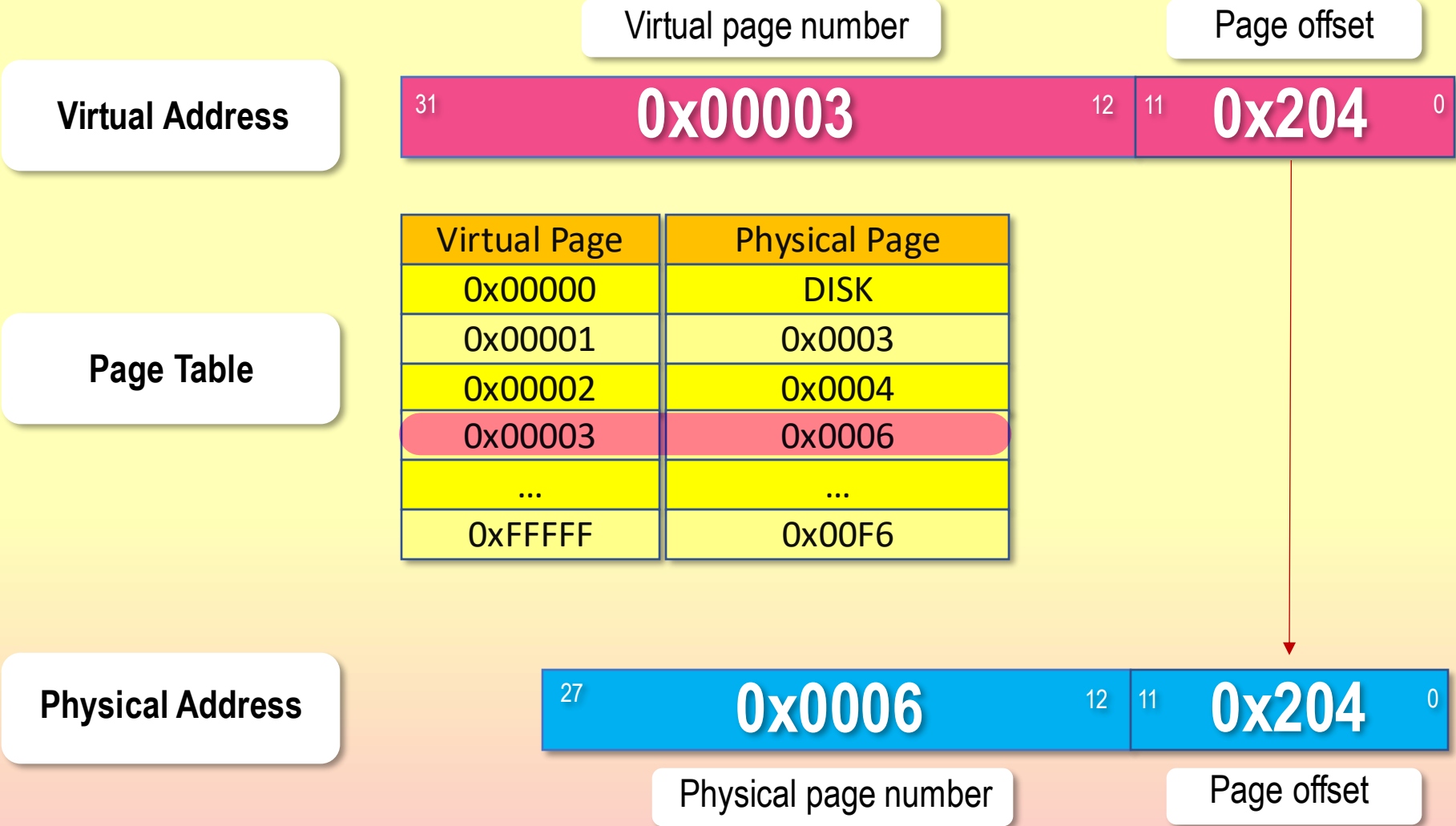
ကိစ္စကိစ္စ Address 000A

0 0
Page 0 was Vir

0 1
Page 1 was Phy

Address Translation

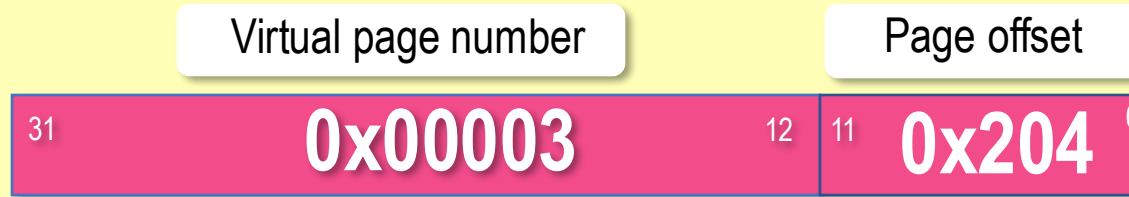
0x00003204



TLB

0x00003204

Virtual Address



TLB

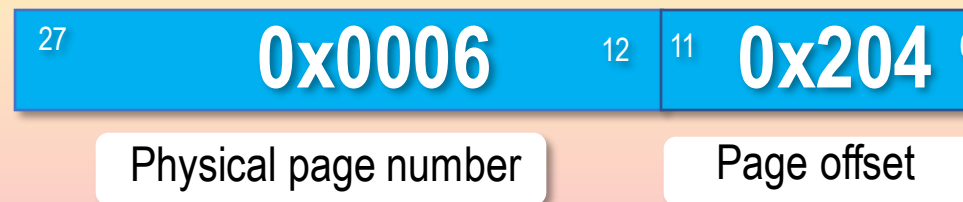
tag	Physical Page
0x00003	0x0006

Cache miss

Memory

Virtual Page	Physical Page
0x00000	DISK
0x00001	0x0003
0x00002	0x0004
0x00003	0x0006
...	...
0xFFFFF	0x00F6

Physical Address



TLB

0x00003208

Virtual Address

Virtual page number

Page offset

31

0x00003

12

11

0x208

0

TLB

tag

Physical Page

0x00003

0x0006

Cache hit

Memory

Virtual Page

Physical Page

0x00000

DISK

0x00001

0x0003

0x00002

0x0004

0x00003

0x0006

...

...

0xFFFFF

0x00F6

Physical Address

27

0x0006

12

11

0x208

0

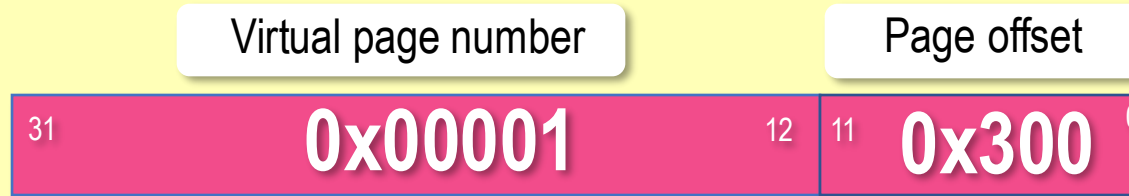
Physical page number

Page offset

TLB

0x00001300

Virtual Address



TLB

tag	Physical Page
0x00003	0x0006
0x00001	0x0003

Cache miss

Memory

Virtual Page	Physical Page
0x00000	DISK
0x00001	0x0003
0x00002	0x0004
0x00003	0x0006
...	...
0xFFFFF	0x00F6

Physical Address

